Heterogeneous Integration for Performance and Scaling

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(*Invited Paper*)

***Abstract*— Moore's law has so far relied on the aggressive scaling of CMOS silicon minimum features of over 1000× for over four decades, and recently, on the adoption of innovative features, such as Cu interconnects, low-*k* dielectrics for interconnects, strained channels, and high-*k* materials for gate dielectrics, resulting in a better power performance, cost per function, and density every generation. This has spawned a vibrant system-on-chip (SoC) approach, where progressively more function has been integrated on a single die. The integration of multiple dies on packages and boards has, however, scaled only modestly by a factor of three to five times. In this paper, we show that with the apparent slowing down of semiconductor scaling and the advent of the Internet of Things, there is a focus on heterogeneous integration and system-level scaling. Packaging is undergoing a transformation that focuses on overall system performance and cost rather than on individual components. We propose ways in which this transformation can evolve to provide a significant value at the system level while providing a significantly lower barrier to entry compared with a chip-based SoC approach that is currently used. This transformation is already under way with 3-D stacking of dies and will evolve to make heterogeneous integration the backbone of sustaining Moore's law in the years ahead.**

***Index Terms*— Circuit sub-systems, flipchip manufacturing, hybrid integrated circuits, integrated circuit technology, manufacturing assembly, memory, electronic industry, Moore's law, multi-chip modules, semiconductor device packaging, system-on-chip, three Dimensional Integration, through silicon via, wafer scale integration.**

I. SYSTEM-ON-CHIP APPROACH

WE BEGIN this paper with a brief review of some key steps in the evolution of silicon technology and the consequences it has had on the development of package and board technologies. In the early days, a few discretely packaged individual bipolar transistors were interconnected by wires to create simple functional circuits. The invention of the monolithic integrated circuit [1] enabled by the Fairchild planar integration process [2] started the trend to functional on-chip integration. IBM's Solid Logic Technology [3] enabled further integration of glass-encapsulated transistors and passives on an insulating ceramic substrate with sintered thick-film wiring—an early example of heterogeneous integration. Further evolution of isolation techniques in the bipolar technology, such as the deep trench isolation, allowed for even tighter integration of devices. Bipolar technology was fast but power hungry, and this in turn led to the development of elaborate packaging techniques for these chips to allow for heat dissipation [4]. Simultaneously, the development of high-quality gate dielectric processes allowed for the development of the MOS technology and later the CMOS technology. The local oxidation of silicon isolation process [5] and its successor the shallow trench isolation [6], combined with Dennard's scaling theory [7], made aggressive scaling possible and thus realized and sustained, until recently, the expectations extrapolated in [8]. Besides the device technology, interconnects are just as important. Metal interconnects were made electromigration resistant [9]; the *RC* delays of the interconnects were also scaled through the development of a viable copper interconnect technology [10], and the damascene process [11] was developed to pattern the scaled interconnects as well as low-*k* dielectrics [12] in the hierarchical wiring system [13], where larger wires are used to connect more distant nodes. All these developments have played an important role in scaling the technology. We highlight these here, because we will argue later that the adoption of similar interconnect approaches is required to scale the system effectively.

Besides technology, the concurrent development of a sophisticated design and test infrastructure has been the key to achieve increased levels of integration and the system-on-chip (SoC) methodology that has driven the microelectronics industry for the last several years. Process capability and process tool capability allow us to define process tolerances, and derive a design manual used to layout devices. The devices themselves are modeled reasonably accurately for the dimensions allowed in the design manual, and these electrical characteristics are used to model the circuits of various complexities. The design methodology starts with these technology attributes and process/physical design kits [14]; design rules that list the constraints that drawn shapes may have, and then with the increased levels of abstraction, allow a designer to start with a functional description of the chip and reduce it to a physical design and layout. Additional tools [15] allow the designer to optimize the layout at different levels of design hierarchy, close timing, estimate power, and statistically ensure chip functionality at some minimum level of performance under allowable process tolerance with a guaranteed reliability level that is derived from an elaborate device and wiring-level reliability models. In addition, there are tools that allow the designer to verify that the chip will be logically correct—a process called logical verification. Finally, the chip must be testable. It is virtually impossible to test modern chips with several billions of circuits for all conditions

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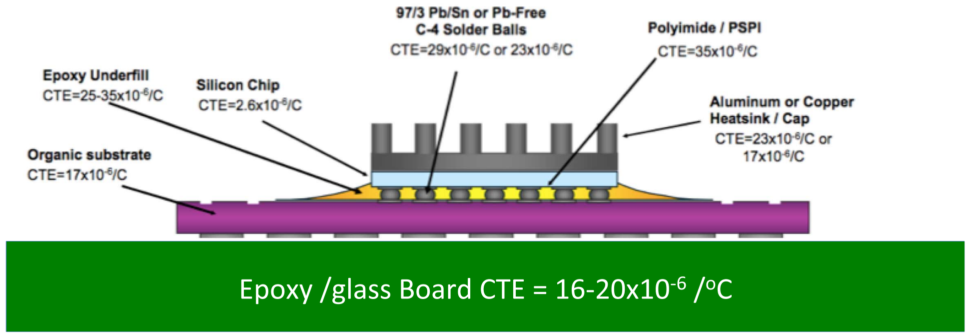

Fig. 1. Anatomy of an organic package.

that will be encountered during its lifetime. An elaborate test methodology has been developed to guarantee the basic functionality by ensuring both connectivity between a large subset of nodes, which guards against manufacturing defects, combined with the measurements of critical delays across the chip, and device performance across the chip to ensure that there are no pathological excursions beyond those assumed by the design system. In addition, for embedded memory and certain critical on-chip functions, full functional test with adequate margins is performed very often through on-chip built-in self-test engines, and there is limited repair capability, especially for memory, through the use of redundancy [16]. Once again, we will argue that this methodology and tool set need to be adapted to a higher level system design.

II. Consequences of SoC Methods on System-Level Integration

The SoC approach described earlier, with the help of scaling, has led to increasingly functionally complex chips, and over time, designers have been able to pull increasingly more function into a single chip and thus reduce the number of chips used in a system. However, a single chip, no matter how much one is able to put on it, does not make a system. In order to build a system, multiple chips [e.g., processors, memory chips, field programmable gate arrays (FPGAs), transceivers, power regulators, and so on] need to be interconnected. Traditionally, this has been done using a printed circuit board (PCB) [17]. A system typically consists of several such PCBs that are connected through a mother board or a backplane using mechanical connectors, whose dimensions are typically of the order of millimeters. Chips are packaged before being mounted on the board. The packages may be attached to the board in a number of ways, including ball grid arrays and land grid arrays, for example. The board is sometimes called the second-level package. The first-level package refers to the die package and is discussed in the following.

After fabrication and test at the wafer level, the dies are singulated and packaged. Fig. 1 shows the anatomy of a typical die attached to a package and a package attached to a board. We show a flip chip package [18] using C4 bumps [19]. There are alternative ways of connecting the die using wire bonds [20]. The packaged chip is often called a module. Traditionally, the package performs several key functions.

1) It protects the chip. The singulated die is considered fragile and sometimes is quite small. The package allows for ease of handling and prevents mechanical damage to the die.
2) It allows the die to be connected to the board and, thus, to the other die by acting as a space transformer. The last wiring levels on the chip are typically called fat wire levels, and can have pitches between 2 and 10 μm. The connections to the board are at pitches that may be a good fraction of millimeters. The package fans out the last wiring level first through the C4s, which are at the typical pitches of 150 μm, and then spaces it out further through multiple package wiring levels to the point where it can be assembled onto the board.
3) Besides signals, power is also delivered to the chip through the package. Power may be delivered to the chip in multiple voltage domains.
4) In the case of moderate (20 W)-to-high (>50 W) power chips, a heat spreader and a heat sink may be attached to the chip with some high conductivity mechanically compliant thermal interface material [21].
5) The package allows the chip to be tested more thoroughly over a wide range of temperatures, and unlike at wafer level, the testing can be accomplished at speed and as close to mission mode as possible with excellent test margins guaranteeing the full functionality.

While the package has performed these useful functions in the past, one must question if today, this is the best way of doing things. Looking at the package architecture in Fig. 1, the package consists of a large number of disparate materials with significantly different coefficients of thermal expansion. This causes enormous stresses on the silicon chip and leads to the so-called chip-package interactions [22] that can cause the low-κ dielectrics, which have lower structural integrity to fail, dies to crack, metal layers to be pulled out, and so on. This is clearly a huge reliability concern. In addition, the package itself can be severely warped to begin with, and this, in turn, adds an additional stress, especially for a large die.

The second drawback of the package is indirect. Fig. 2 shows the interconnect pitch as a function of (vertical and lateral) distance from the gate of a CMOS chip. While it is true that the package connects the chip to the rest of the system, it does so very inefficiently. Consider that contacted gate pitches in the 14-nm node are about 40 nm, through the hierarchical wiring system, we increase that pitch up step-by-step until, at the upper most wiring level, it is a few micrometers. The C4 bumps then increase this pitch to

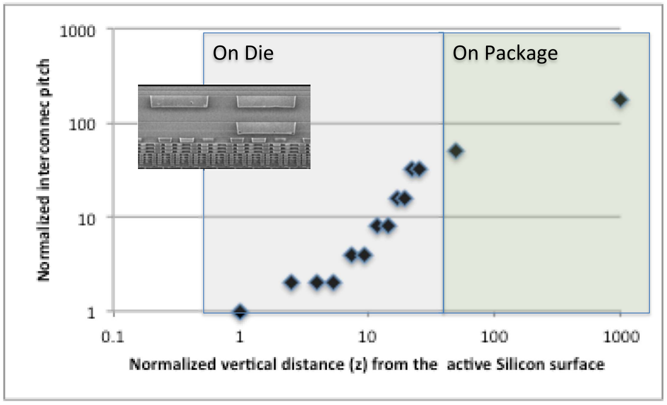

Fig. 2. Normalized interconnect pitch as a function of vertical distance from the active die surface for a typical 45-nm SoC. In a hierarchical wiring system, lower tight-pitch levels are used for local interconnects, while the higher thicker and lower resistance levels are used for longer connections. For reference, M1 pitch is typically about 144 nm, bump pitches are at 150–200 μm, and BGA pitches are at 400–600 μm. Board trace pitches are typically in the 50–200-μm regime. Inset: SEM micrograph of some of these levels.

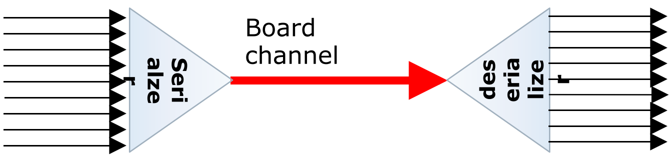

Fig. 3. Limited wiring at relatively large pitch on the board requires us to serialize signals and transmit them at high speed to other chips and deserialize them there.

about 150 μm. The BGA connections to the board take this further to several hundred micrometers. Trace pitches on the board are between 30 and 100 μm, and one can have several levels of wiring to ensure wirability. In fact, there is a tradeoff between the trace pitch and the number of wiring levels. The traces may terminate in either a mechanical connector with mechanically aligned pins of macroscopic dimensions or at another chip, where the connections are fanned-in through progressive space transformation in reverse.

This fan-out and fan-in process has important consequences in chip design. First, as we pointed out earlier, these dimensions have not scaled appreciably in the last few decades. The large bump pitches limit the number of I/Os that can escape a chip. Typically, these I/Os are arranged near the edges of the chip closest to the chip they are trying to communicate with. For example, in a typical microprocessor chip, we may have about 10 000 bumps of which barely 1000 will be allocated to I/Os and the rest will be allocated to power and ground. Since the bump pitch and the length of the edge of the chip limit the number of I/Os, chip designers have come up with many ways to boost the bandwidth (BW) of the chip. The classical way of doing this is to serialize, transmit, and then deserialize the signals [23]. This is shown schematically in Fig. 3. However, with increasing demands on the BW between the chips and the inability to increase the number of physical connections between the chips, the serial links need to operate individually at higher and higher data rates. These higher rates mean higher frequency signals carried by the traces on the board and significantly larger noise levels and cross talk between adjacent channels. There are two consequences of this trend.

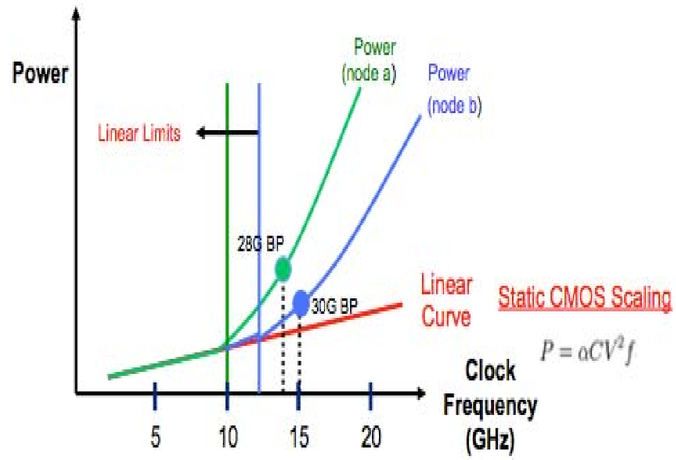

Fig. 4. Exponential increase in power as a function of data rate (adapted from [23] and [24]). Node b is smaller than node a. Also shown are the linear power limits of static logic circuits.

1) The power to transmit higher frequency signals through SerDes goes up exponentially with data rate [24], as shown in Fig. 4.
2) The SerDes circuits themselves become more complex to design and take up more area [25].

While the details of why this is so are beyond the scope of this paper, the reader is referred to several excellent tutorials on the subject [26]. As a consequence, SerDes design IP has become a key-differentiating factor in SoC implementation, and several somewhat independent and noninteroperable protocols have been developed by different companies to differentiate their products. Even so, modern SoCs may sometime devote almost 25% of their area to SerDes, and in some cases, an even greater fraction of chip power is allocated to SerDes function. Interestingly, significant component SerDes circuits are analog, and unfortunately, these have not scaled as well as digital logic, further impacting the efficiency of these designs. In addition to drive the high wiring loads on the board with good signal-to-noise ratios, SerDes usually operate at 30%–50% higher voltages than the chip supply voltage further compounding the power problem. Furthermore, since signaling is done with transmission lines, one needs to ensure impedance matching—usually through dissipative terminations and low resistance ground returns. The high-power usage of SerDes makes local decoupling essential, and this too adds to the area.

The memory subsystem is particularly egregious in this regard. Memory uses a moderately parallel interface, but it is limited by pin-outs and this thus runs at a very high frequency. Though technically not a SerDes, it shares many of the drawbacks of SerDes, and at the board-level double data rate, channels are designed as terminated transmission lines. In fact, memory power is a very significant fraction of system power. At the system level, it accounts for 30%–40% of the total power of which data transfer to the processor chips dominate. In fact, for the most large-scale systems, within system communication may use 30%–40% of the power budget, as shown in Fig. 5.

Finally, we must address board-level latency. While it is possible to address the problem of intrasystem BW and

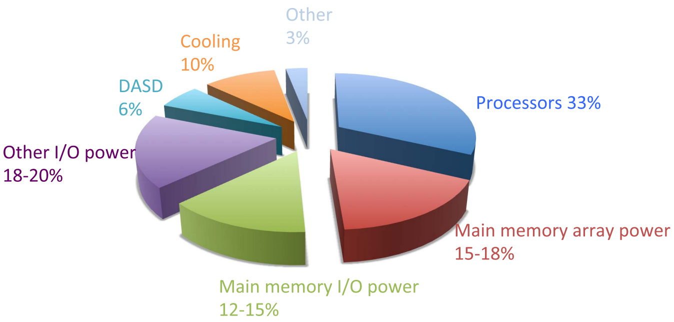

Fig. 5. Approximate breakdown of power in a mid performance system. I/O power including memory power accounts for more than a third of total system active power.

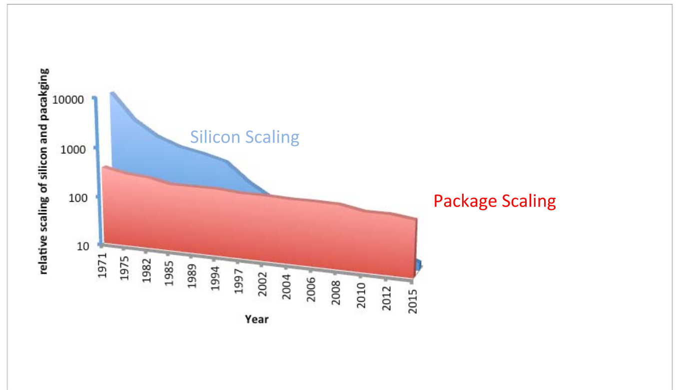

Fig. 6. Relative scaling over time for silicon and packaging features. Silicon has scaled by 1000×, while packaging features have scaled by a factor three to five times. This shows bump pitch, but other packaging features behave similarly. In the late sixties, minimum Si features were about 15 μm, and bump pitch was about 400 μm. Today, they are 15 nm and 150 μm, respectively.

data rate with the expenditure of chip area and power, the raw time to transmit and receive data (latency) is ultimately related to various delay components on the data path, including the length of the data path as well as the time to process data at the transmitter and the receiver, and is often a limiting factor in system performance. At a higher level, latency determines overall system performance, and it is important to reduce latency while increasing system BW. On the transmission line, data travels at speeds approaching the speed of light (in the dielectric material used on the board). However, the SerDes process as well as the other address decode operations also add latency. For a typical processor board, the overall latency between the processor (at one end of the board) and the hard disk drive (HDD) can be as much as a few milliseconds. One can reduce this considerably by employing solid-state drives, at least for read operations. It is possible to reduce this latency by about two orders of magnitude by packing devices closer, increasing wiring connections between the die and the appropriate change in signaling protocols.

III. Limitations of the SoC Approach

The SoC approach while immensely successful so far is straining the limits for several reasons.

1) Technology scaling is saturating. We are no longer getting the 2× area improvement nor the 2× performance improvement per node. The cost per circuit is not reducing, and the development cost per node is increasing rapidly.
2) As SoCs get more complex, besides the cost of technology, design costs are also increasing. This includes design tools, layout with restrictive design rules, verification, mask costs, prototype costs, and test costs. This higher barrier to entry stifles innovation and we are seeing a consolidation of both fabrication companies as well as Fabless design companies.
3) New applications are driving systems to be more heterogeneous. This is a key point. In the compute area, the traditional von Neumann model with a well-defined memory hierarchy is breaking down—computation is becoming more data centric and this means the traditional linear configuration of processor, different levels of cache all the way to HDDs need to become more memory centric. In the mobile area, the so-called Internet of Things (IoT) is requiring more heterogeneous systems that include a variety of sensors, micro electro mechanical systems (MEMS) devices, and energy storage and power distribution devices. While scaling has worked well so far for digital devices, it has not been as spectacular for analog devices, for example. Thus, an SoC with 10% analog content in a 130-nm technology can have as much as 40%–50% analog content in the 45-nm node.
4) These newer nodes being more complex take a longer time for yield to ramp up and stabilize, especially for large SoC dies. Often terminal yields do not reach historically predicted levels. This means that there is an additional penalty in chip cost.

IV. Approaches to System Scaling and Heterogeneous Integration

To address this issue, one must take a somewhat holistic approach to system scaling. In Fig. 6, we show minimum silicon feature scaling with time and compare that with scaling in package scaling—whether it be bump pitch, BGA pitch, or trace pitch, package features have not scaled to the same extent as silicon features. Silicon has scaled by over a factor of 1000 in the last 50 years, while packages and boards have scaled by at most a factor of 5, as shown in Fig. 6. To achieve a holistic scaling, we need to consider how we can scale the overall footprint of the system. In the following, we suggest a systematic approach to do so. Some of these transformations are underway already, but we need to accelerate the pace considerably.

1) *Reduce the Die Footprint:* Classical scaling has been reducing the die footprint but increasing die content has more than compensated for this. As a result, die sizes have not been decreasing significantly. This is true for both logic and memory dies.
2) *3-D Integration:* Stacking die is a very efficient way of reducing die footprint. This issue of the IEEE components, packaging, and manufacturing technology Transactions has several papers devoted to this subject. This topic merits more discussion and we will return to it later.

3) *Eliminating the Package:* We had argued earlier that the value proposition of the package was significantly reduced and we suggest here that eliminating the package and directly bonding multiple bare dies to an interconnect fabric (IF) will significantly reduce the overall system footprint. The first steps in this direction have been taken with silicon interposers (discussed in detail in Section V). However, we argue that this has not gone far enough. While we have been able to integrate a few die on an interposer, we have done so with even more complexity—adding an extra level to the overall package. However, interposers have been able to demonstrate tight interconnections between die, heterogeneous integration, as well as power management and distribution, and heat sinking [27] as well as the advantages of going to smaller die [28]. What we are proposing here is to go a step further and transform the interposer into a full fledged board. We will elaborate on this proposal in a later section.

V. 3-D Integration

While the concept of stacking die is not new, package-on-package [29] technology has been practiced quite extensively, for example; with 3-D integration, there has been a spurt of innovation in recent years that has made footprint reduction by stacking bare thin die a reality. In addition, silicon interposer technology may be considered a subset of 3-D integration where the bottom die is larger and supports multiple top dies and where the bottom die is passive in that here are no active devices. Several innovations have played a key role in making this technology possible. These have been reviewed elsewhere [30] and we only highlight the main points:

1) high aspect ratio through-silicon vias (TSVs) using the deep reactive ion etching technique called the Bosch etch [31] developed by the micromachining community;
2) high aspect ratio high-quality dielectrics to insulate the TSVs;
3) conformal copper barrier (TaN or TiN) and copper seed deposition;
4) high aspect ratio bottoms-up plating technology with elaborate microstructure control;
5) integration schemes and methods to manage copper pumping and creep by microstructure control and thermal treatment (pumping occurs because of the large thermal expansion coefficient mismatch between silicon and copper);
6) thin wafer handling techniques;
7) assembly techniques that incorporate copper pillar and microbumps with pitches as small as 40 μm;
8) methods to control thin die warpage;
9) understanding the impact of TSV stress and formulating keep-out zones (KOZs) to minimize device perturbations due to stress.

However, by far, the implementation and acceptance of 3-D integration have depended on finding an application that is consistent with the cost of a nascent 3-D process where volumes have not kicked-in to enable large volume applications and one that provides genuine value. So far, there have been two classes of applications.

A. Memory Stacking

We showed, in Fig. 5, that memory power can account for one third of the system power. Much of that is signaling power. Moreover, as computation becomes more data centric, memory activity is expected to rise and this fraction will only increase, and in addition, the number of memory die that needs to be integrated will increase, in turn increasing the size, power, and cost of the systems. Memory manufacturers have taken three main approaches to address this problem and these are in moderate production already.

1) Simple memory die stacking with no architectural changes, mainly a form factor improvement.
2) *Hybrid Memory Cube [32]:* This is an example of a master-slave [33] approach where a logic die at the bottom of the stack provides addressing, control, test and repair function, as well as SerDes communication to a processor die. Communicating within the stack is done through TSVs with low latency and high BW. The logic die and up to eight memory stacked dies are packaged in a conventional organic package and assembled by the conventional techniques on a PC board. We consider this a half step as we have not completely eliminated the SerDes although reduced its use considerably. With 16 channels or lanes, each with a nominal data rate of 10 Gb/s, the total raw BW can reach 40 Gb/s (20 Gb/s each for transmit and receive). It is estimated that at similar data rates and memory capacity to the conventional memory dual in-line memory modules, hybrid memory cube (HMC) reduces memory footprint by $\sim$90%, memory power by 30%–35%, and the number of active signals by over 80%. These results are impressive enough to make the HMC a very compelling proposition.
3) *High BW Memory (HBM) [34]:* This too is in the master-slave class but with some important differences. HBM leverages a large number of I/O pins—1024; each pin has a smaller BW per pin of 2 Gb/s, but, by increasing the number of I/Os from 16 to 1024, the HBM is able to get BW of 128 Gb/s per stack. The caveat is that since the number of wires between the die and the processor is large, and also at a fine pitch of 50 μm, HBMs and the processor associated with them need to be mounted on silicon interposer.

All three of these approaches have established commercial applications. The simply stacked DRAM is used in midrange servers used in data centers. Multiple companies use HMC for high-performance computing and HBM recently made its debut in graphics processors used in gaming systems. Fig. 7(a) shows an HMC stack, and Fig. 7(b) shows four HBM stacks integrated with an AMD graphics processor [35]. In addition, all these three applications stack dies face (device side)-to-back (f–b)—with the face of the bottom-most die attached to the laminate. The advantage of this method is that it allows us to stack multiple dies sequentially. The disadvantage is that all connections need to go through TSVs.

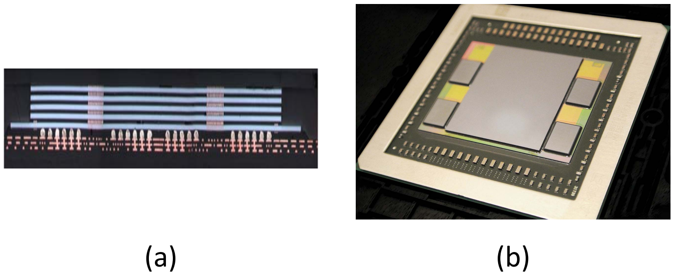

Fig. 7. (a) HMC chip stack showing the lower logic die with four stacked memory die. Courtesy Micron Technology and IBM. (b) Interposer application with stacked four HBM die and an AMD processor Courtesy AMD.

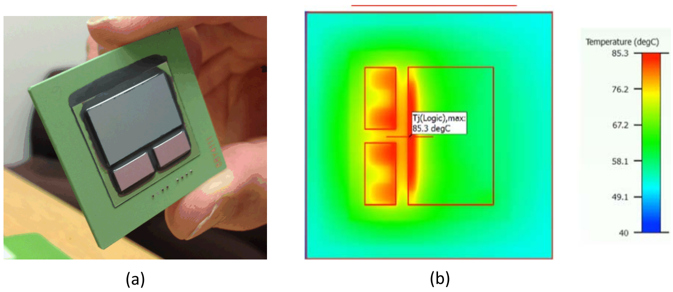

Fig. 8. (a) Example of heterogeneous integration of analog and mixed signal SiGe transceiver chips and an ASIC processor. The chips were fabricated with optimized processes separately in different fabs and integrated on an Si interposer. (b) Thermal profiles of the interposer product, as shown in Fig. 7(a). Most of the activity power is dissipated in the high-speed links at the periphery of the die and is extremely nonuniform. Peak temperatures may increase by 40 °C—a good reason to minimize the use of high-speed serial links.

Since TSVs take up die real estate and disrupt design, this can be a severe penalty when two adjacent dies need to communicate.

B. Heterogeneous Integration

We pointed out earlier that as SoCs get more complex, they get more complex to design and fabricate. Two examples that bring this out are the integration of high-speed analog transceivers with processors and the integration of memory and processor. We have already described the interposer application of HBM and graphics processor. Another similar example is the integration of two 90-nm SiGe BiCMOS transceivers with a 45-nm application specific integrated circuit (ASIC), as shown in Fig. 8(a). This integration could never be accomplished in a monolithic technology, as the processes are incompatible. Yet, this interposer implementation allows for seamless integration with a interdie BW that exceeds 2 Tb/s. Fig. 8(b) shows the thermal profiles of the die. It shows most of the heat is generated by the high-speed connections between the die.

Another example is the integration of cache on processor. Here, the requirements go beyond BW, and latency is an even more stringent criterion. The latencies of a <2.5–3 ns (or even less) required for cache access in high-performance systems cannot be obtained by the aforementioned memory solutions for two reasons: 1) the intrinsic latencies of the commodity DRAM products are large and 2) the electrical distance between the memory stack and the processor is of the order of 10 ns. On-chip large dense memory [36],

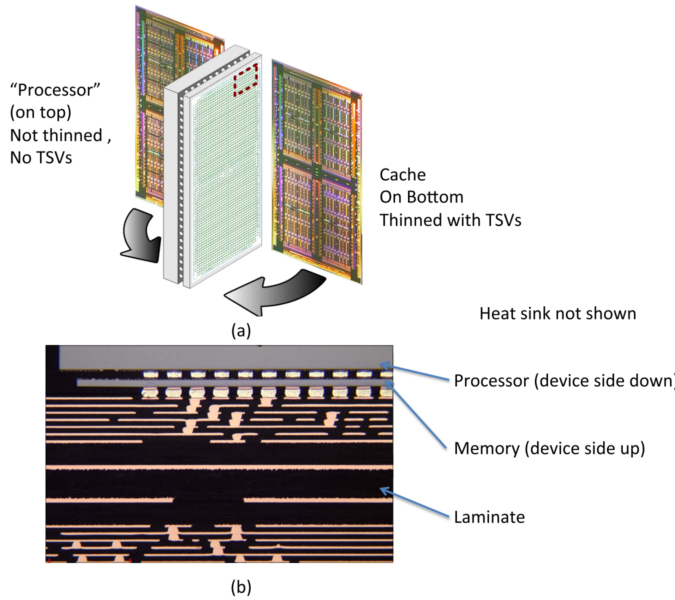

Fig. 9. (a) Schematic of face-face Processor and cache assembly. (b) Optical cross-section of an assembled face to face 3-D stack.

such as embedded DRAM, is ideally suited for this application, but as the number of cores increase, the memory size has been growing to the point that die sizes have increased to unreasonable sizes ($\sim$800 mm^2). In addition, while fast DRAMs do in fact need fast logic technology, they do not require the large logic device menu nor the large number of metal levels for wiring. Conversely, the logic does not need the complexities of capacitor formation. In addition, yields of the cache and processor are coupled, each limiting the other. A simple way of alleviating this problem is to stack separately fabricated optimized cache and processor die in a face-to-face (f–f) configuration, as shown in Fig. 9 [37]. The processor die is mounted on the top of the memory die and connects to it directly through microbumps. This ensures two things: 1) a very large number of connections ($>$0.5 million/cm^2 at 40-μm pitch) and 2) extremely low latency between the two die ($<$0.3 ns each way). This communication is done without SerDes. Both allow for the cache to have performance and power comparable to an on-chip cache. Other heterogeneous applications for power delivery, regulation and management, and RF integration are also high leverage applications.

It has been argued that 3-D integration has been slow to gain commercial acceptance. This is a somewhat unjustified criticism. If one looks at other game changing technology transformations (described in the beginning of this paper), such as the introduction of new materials in silicon technology, including silicides, copper, low-k dielectrics, lead free solder, and new device concepts, such as the bipolar to CMOS transition for digital applications and Si–Ge devices, the acceptance of 3-D integration has been comparable. Dramatic changes in the packaging area have taken even longer to get established: flip chip bonding as well as the transition from ceramic packages to organic packages have taken even longer. All of these technologies have had challenges—both technical and economic, but 3-D integration is more than a materials introduction—it is the beginning of new way of doing things and an important, albeit first step in how we will integrate electronic systems of the future and requires architectural accommodations as well.

VI. Thermoelectromechanical Codesign

A 3-D chip stack is a complex design, and there needs to be a high level of codesign of the chips in the stack. Mechanical considerations such as stress due to TSVs can affect device performance and must be modeled and if warranted need to be extracted from layout, schematics, and device models. Thin dies can warp, and the warpage can affect assembly, especially for large die. This warpage is dependent on the details of the layout, the positioning of TSVs and other fat wire pattern densities. Today, there is no systematic way of integrating these three components of design. Therefore, most designs in the market are iterated in a custom manner. However, for 3-D stacking to become more widespread, this *ad hoc* custom methodology needs to be formalized and incorporated into the standard EDA environments.

A. Thermal Considerations

A technical challenge that will limit the introduction of 3-D stacking and 3-D integration in general is thermal management. In a chip, heat is generated at the device and also to a lesser extent in the wiring. In general, chips are mounted face down on the package—which, typically, is an organic laminate that is not a good heat spreader or sink. Furthermore, typical modern chips may have several micrometers of back-end oxide, which are thermally insulating. Therefore, about 90% of the heat is dissipated from the back of the die through silicon. Heat dissipation is managed through the use of heat spreaders that spread heat away from the chip laterally and then dissipated through heat sinks that are cooled either by ambient air or forced air-cooling. In the case of high-power chips, fluidic and phase change cooling is also employed.

In the case of stacked memory die, this problem is somewhat alleviated by the intrinsically lower power of the memory die. However, in the f–b configuration, we have, in this case, the logic controller die with the I/Os is necessarily close to the laminate, and while it produces the most heat, it is also the farthest away from the heat sink. One way out is to ensure that the logic die is somewhat larger, as shown in Fig. 7(a). Most of the logic power is generated in the I/Os, and this is usually at the periphery of the chip. Using a metal lid with a lip, it is possible to heat sink the edge of the exposed bottom die somewhat better. Therefore, it is important to minimize this issue by ensuring that the logic chip power is kept within acceptable limits (typically $<$10–15 W) if we do not want an elaborate cooling solution. Another approach is to use DRAM-based logic, which is lower power but also lower performance. This is the current approach used in the HBM, but it is not clear whether the lower performance DRAM transistors can perform the logic functions satisfactorily in the next-generation HBMs. In general, it is fair to say that when contemplating multiple die stacks, low power is a must. Even with a heat sink on the upper most die (away from the laminate), heat generated at the lower strata must pass through the C4 bumps/Cu pillars (the under-fill is assumed to be nonconducting), the back end of the die mentioned earlier and the silicon itself. As a result, we will expect temperatures to gradually increase from the top (close to the heat sink) to the bottom. Moreover, since the logic chip runs the hottest and is heat-sunk the least, it is important to ensure that any temperature rises are within the operating specifications of the dies.

In the case of the processor/cache assemblies discussed earlier, the processor is close to the heat sink and to the first order experiences the same thermal environment as a 2-D processor. The cache chip below is lower power—about 10–15 W/cm^2 (caches are dominated by logic) and runs cooler than the logic chip and so an interesting thing happens: for the processor die, heat flows both up to the heat sink and also down to the cache chip actually raising its temperature beyond what one might expect for a standalone cache chip. Once again, careful precautions must be taken to ensure that this temperature rise is within acceptable limits (a few degree Celsius at most).

These thermal considerations can be designed-in by codesign to ensure that during the operation of the stack, temperatures are kept under control by monitoring temperature at various points in the stack and using firmware to manage activity. Even so, we certainly do not expect to see memory die stacks in excess of 16 strata as even a 1°–2° increase per strata would cause the bottom die temperature to increase by about 16 °C–32 °C—A 8°–16° rise in temperature can be managed through design, and a 4–8-high stack with a logic die will be the norm. In the case of high-power processors, thermal and power distribution considerations will make anything other than a two high stack difficult. Clearly, thermal management in general will limit the applications of 3-D stacking and a case can be made for a concerted effort to address this issue economically.

A case may be made to address this problem in a more systematic manner, including if possible, the use of fluidic cooling and even heatpipes. Such solutions while feasible [38] will not be economical but for the highest end applications.

B. Power Delivery Considerations

Power delivery is an important consideration. In both configurations discussed earlier (f–f and f–b), all power to upper strata will need to pass-through the lower strata through TSVs delivery to ensure that IR drops do not cause voltage reduction and consequent performance reduction. For processor dies, this is a challenge, as the power grid needs to be extremely uniform and delivered with no more than a few millivolts of IR drop. In 2-D processor die, approximately 90% of the C4s are allocated to power delivery. If we have a single TSV for each power C4, we estimate about 2000 TSVs/cm^2 allocated for this purpose. At typical TSV dimensions, and accounting for generous KOZs, this works out to less than 1% burden. However, the TSV seriously disrupts the design, and one needs to design around the TSVs. In the worst case scenarios, this can cause up to $\sim$20% inefficiency in design [39]. Similar issues, though to a much lesser extent, affect the stacking of lower power die. One approach is fan-in power from multiple C4s attached to the laminate side and to deliver them through the thinned die through a single TSV and then use the final thick metal layer of the lower die to distribute the power uniformly to the upper die through microbumps. Using this approach and the so-called TSV farms (where the KOZ burden is minimized) and other design optimizations, one can reduce the TSV design overhead penalty to $\sim$5%. Nevertheless, in certain cases, this might not be possible and the density advantage of 3-D stacking may be eroded. Automated design tools today do not automate this optimization; though it is expected they will as 3-D gets more established.

C. Signal Integrity Considerations

Signals within the 3-D stack propagate through TSVs, and for all intents and purposes, these act as wire loads with an important difference: the resistance is low, because TSVs are large in size, but the capacitance is high. At frequencies of about 1 GHz, the RC skew is significantly less than picoseconds. However, as in the case of power distribution, in the f–f case, signals need to propagate laterally along the chip until they reach a TSV. This lateral propagation needs to be accounted for signals that are escaping the stack to the laminate.

Much of 3-D design today is custom and not automated. To be used widely, the custom knowledge needs to be systematized and formalized, and design methodologies need to be incorporated into the standard design tools. However, much of memory design today is custom and thus we expect 3-D applications with large memory content to be dominated by custom approaches.

VII. Future Directions

In Section IV, we enumerated the approaches, one could take to reduce system footprint, and in Section V, we described 3-D integration including interposers as one component of this to reduce die footprint. However, we feel this is only a first step and much work remains to be done. We mentioned that systems are getting more heterogeneous, and components cannot be coprocessed (because of incompatible materials—say Si and GaN) or have different scales (large-area sensors versus Si devices), for example. There have been proposals that introduce the concept of a system on package [40]. However, progress on this front has been limited. The primary reason is that while these approaches have reduced the form factor, they have not realized the benefits of enhanced system performance. This is because we have not improved the interconnect pitch and lateral dimensions of the system and have remained more or less the same. In addition, the package remains small by board standards, and so we are limited to the number of components we can integrate.

Even so, packaging is undergoing a renaissance of sorts. There has been a greater adoption of siliconlike processing techniques as evidenced by the wafer level fan-out process [41] now practiced by several companies in several forms. Methods such as these leverage the rich experience of silicon technology by reconstituting a wafer with good die and employing siliconlike processing to process finer features and fan-out wiring leading to more compact packages with higher performance and lower losses.

Another approach has been the use of selective fine pitch features on multichip packages using the so-called embedded multichip interconnect bridge [42], which uses strategically placed silicon slivers with built-in tight-pitch wiring that is embedded in the package. The claim here is that by limiting the use of tight interconnects, one can achieve good performance at lower cost. However, the challenge of connecting the die to the bridge at fine pitch remains. For future SoC replacement applications discussed in the following, it may not be possible to limit the availability of fine pitch wires to just the bridge areas in which case one must increase the area covered by bridgelike connections. The consequence of this is discussed in the following.

VIII. Silicon—New Packaging Material [43]

The important point here is the increasing use of silicon processing and silicon in packaging. We had questioned earlier the value of the package in modern systems. Here, we elaborate on this concept. We propose that we replace the board material with a silicon board. For the moment, let us assume that it is a silicon wafer on which we have processed several levels of fine pitch wiring with the top-most wiring level matching the fat wire levels with landing pads of similar dimensions that can receive connections from die that can be precision aligned. We call this wafer the silicon IF (Si-IF). Small die can be assembled with precision alignment (0.5-μm overlay) onto the Si-IF using the state-of-the-art alignment tools. Electrical connections between the rigid flat die and the Si-IF can be made by thermal compression bonding. While this technology at this fine scale is not available today, we do believe it possible in the near future. Two features of this approach make it more feasible—the use of small die (a few millimeters on a side) and the fact that both the die and IF are made of relatively thick silicon and are flat and with matched CTEs.

We believe there are several factors in the evolution of systems that make this possible and attractive. First, consider computation, which is becoming more data centric. Cache sizes cannot grow indefinitely and it will be more productive to bring main memory closer to the processing unit, so that the entire working set will be local. The availability of stacked memory (of all flavors) and high BW access makes this a logical next step. The scheme we have proposed will also reduce latency to levels that can approach that of an on-chip cache within a factor of 4 or so. By interleaving cores (small die) and memory of comparable footprint, we can achieve a much more balanced and distributed system that will yield and perform better. Furthermore, computing is evolving to a more heterogeneous architecture with a combination of special purpose processors, accelerators, and FPGAs which make this Si-IF integration scheme very attractive, since one can synthesize such systems from a variety of off-the-shelf components. In the case of mobile systems and the so-called IoT, heterogeneity is, indeed, the name of the game. One can now integrate analog components, sensors, MEMS, batteries, supercapacitors, and so on as needed. Overall, we expect that this approach will allow us to reduce overall board footprint by about a factor of 3–2.

The last several years have seen the proliferation of larger chips with more heterogeneous IP being integrated on a single die. We argued earlier that this approach was not sustainable and that we need to develop a viable alternative. The Si-IF approach we have proposed is one such option. Moreover, one can conceive of integrating a small number of active functions as well on the Si-IF. These include power management, repeaters, buffers, and even gate arrays. However, there needs to be several concomitant developments for this approach to become successful, widely adopted and economical. We outline some of these in the following.

1) *Integrated Design System:* Today, chips, packages, and boards are all designed separately and almost independently. This will need to become a lot more integrated. In addition, while, today, we do electrical, thermal, and mechanical design more or less independently, these three views of the system will also have been integrated. Such a system will require significantly more understanding of the interactions between these three views and the development of a significantly more sophisticated set of tools.
2) *New Design for Test and Repair Methodologies:* As rework is no longer an option, and die-level testing will be limited in scope, the component chips will have to test themselves to a large extent. While technologies to do this exist, they will need to be adapted to bare die. Furthermore, the design methodology will need to comprehend failure modes and be resilient to them as well, either through redundancy or extreme fault tolerance driven by system architecture. This too is an enormous challenge. In some cases, fine pitch bare die test equipment and methodologies will need to be developed.
3) *Interface Standardization:* Our approach allows us to have a large number of interdie connections and this allows us to parallelize the connections and have simpler interfaces. However, this approach needs to be adopted universally. Such adoption requires a mindset change and will not happen easily. However, we believe that the standardization of slower and easier to build parallel interfaces is more easily achieved than serial interfaces.
4) *Power Delivery and Thermal Management:* In the case of high-end systems, one would need to deliver $\sim$1 kW of power at different voltages. This will require integrated power management techniques, and the use of features, such as Hi-Q inductors, buck convertors, and power switches, than can either be components attached to the IF or integrated into the Si-IF. Removing the generated heat is another challenge. One mitigating factor in the use of the Si-IF is that the silicon itself is a good thermal conductor and can be an integral part of the heat-sinking solution. The die themselves may have integrated heat sinks made, for example, with silicon fluidic channels or micromachined fins. These approaches need to be comprehended in the integrated design system discussed earlier. Methods that are not cost effective at the die level may become cost effective at the system level.
5) *Structural Properties of Silicon:* While the ability to process silicon as an IF is second to none, silicon is quite brittle and thus care must be exercised in wafer handling. Fortunately, silicon-processing equipment has evolved to accommodate this brittleness and wafer breakage in the Fab is a negligible component of yield loss as long as the wafers are thick enough. We expect these techniques can be extended to IF processing as well. After assembly, the additional silicon is expected help in strength but care in handling will be needed. In addition, there are reports [44] that surface modification nanoindentation can reduce brittleness. Even though silicon is brittle, it does demonstrate high fracture strength (unlike glass). In addition, additives, such as oxygen and nitrogen, can modulate strength.
6) *Cost:* It has sometimes been argued that silicon is expensive and organic materials would be more cost effective. If we need fine pitch interconnects, then in practice, material cost will be about 10% of the finished product cost. Processing the fine pitch interconnects dominates the cost. In fact, it will be very expensive to fabricate fine pitch (sub-10-μm pitch) interconnects on organic substrates; while fully depreciated silicon Fabs can do this easily and more cost effectively on silicon. There are additional benefits of silicon, such as integrated passives and active IFs. The silicon solar cell substrate suppliers have developed the so-called metallurgical grade Si that is extremely competitive with organic substrates, but as we mentioned earlier, the cost of fine pitch wiring will dominate. Nevertheless, a rigorous cost–benefit analysis that includes value addition, such as time to market, ease of design, and heterogeneity to increase value and differentiation, must also be considered.

While the challenges are enormous, so too are the payoffs. When compared with the challenges and costs of shrinking minimum features on a die, we believe the value proposition of what we have proposed here is solid.

IX. Conclusion

Packaging needs to be recast in terms of value-addition system integration. It is no longer a way to protect die but rather a way to interconnect them and the value of this integration scheme will depend on more electrically tight connections, simpler die-to-die connection protocols that achieve higher BW with lower latency and at lower power. We have taken the first steps in 3-D integration where die stacks have demonstrated these very attributes. However, this is but the first step. We have proposed a scheme based on a silicon IF that will allow us to integrate a heterogeneous collection of dies, manufactured in different processes, and subscribing to a somewhat universal protocol. While several capabilities still need to be developed to make this vision a reality, the first steps have already been taken through the use of 3-D stacking and interposer technology. The adoption of silicon processing in packaging is another key development that will enable this transformation.

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